

# Python Programming

## Unit 06 – Lecture 03: Pandas Basics (Series and DataFrame)

Tofik Ali

School of Computer Science, UPES Dehradun

February 20, 2026

Repository: <https://github.com/tali7c/Python-Programming>

# Quick Links

Core Concepts

Demo

Interactive

Summary

# Agenda

1 Core Concepts

2 Demo

3 Interactive

4 Summary

# Learning Outcomes

- Explain what Pandas is used for

# Learning Outcomes

- Explain what Pandas is used for
- Create and manipulate Series and DataFrames

# Learning Outcomes

- Explain what Pandas is used for
- Create and manipulate Series and DataFrames
- Use `head()`, `tail()`, column selection, and basic iteration

# Learning Outcomes

- Explain what Pandas is used for
- Create and manipulate Series and DataFrames
- Use `head()`, `tail()`, column selection, and basic iteration
- Perform vectorized operations on columns

# Why Pandas?

- Works with tabular data (like Excel/CSV)



# Why Pandas?

- Works with tabular data (like Excel/CSV)
- Powerful cleaning and transformation functions

# Why Pandas?

- Works with tabular data (like Excel/CSV)
- Powerful cleaning and transformation functions
- Easy grouping/aggregation and missing value handling

# Series and DataFrame

- **Series:** 1D labeled array

# Series and DataFrame

- **Series:** 1D labeled array
- **DataFrame:** 2D table with labeled columns

# Create a DataFrame

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Asha", "Bilal", "Charu"],
    "marks": [88, 76, 91]
})
```

# Useful Methods

- `df.head(n)`, `df.tail(n)`

# Useful Methods

- `df.head(n), df.tail(n)`
- `df.info(), df.describe()`

# Useful Methods

- `df.head(n), df.tail(n)`
- `df.info(), df.describe()`
- Select a column: `df["marks"]`



# Add/Delete Columns

```
df["cgpa"] = df["marks"] / 10  
del df["cgpa"]
```

# Vectorized Operations

- Column math applies element-wise (fast)

```
df["marks_plus_5"] = df["marks"] + 5  
df["passed"] = df["marks"] >= 50
```

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:
  - creating Series/DataFrame

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:
  - creating Series/DataFrame
  - head/tail

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:
  - creating Series/DataFrame
  - head/tail
  - selecting and adding columns

# Demo: Series + DataFrame Operations

- File: `demo/pandas_basics_demo.py`
- Shows:
  - creating Series/DataFrame
  - head/tail
  - selecting and adding columns
  - basic boolean filtering



# Checkpoint 1

**Question:** What is the difference between Series and DataFrame?

## Checkpoint 2

**Question:** How do you select the "marks" column from a DataFrame?

# Think-Pair-Share

Discuss:

- Why are vectorized operations preferred over loops in Pandas?

# Key Takeaways

- Pandas simplifies tabular data handling

# Key Takeaways

- Pandas simplifies tabular data handling
- Series is 1D; DataFrame is 2D

# Key Takeaways

- Pandas simplifies tabular data handling
- Series is 1D; DataFrame is 2D
- Column operations are vectorized and efficient

# Exit Question

Write one line to create a new column "cgpa" from "marks".